MAHATMA GANDHI UNIVERSITY,KOTTAYAM

FIRST SEMESTER MGU-UGP(HONOURS)

REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course-MG1DSCNBC100

FUNDAMENTALS OF NUTRITIONAL BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

QP CODE: 24900197

Duration: 2 Hours Maximum Marks :50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

Part A
Multiple Choice Questions
Answer all questions
Each question carries 1 mark

1Glycogen (1 mark)
2.form of storage carbohydrate in plants (1 mark)
3.Myoglobulin (1 mark)
4.Cysteine (1 mark)
5.Vitamin E (1 mark)
6.Calcium absorption (1 mark)

Part B
Short Answer Questions
Answer any two questions
Each question carries two marks

7. In research, Diagnosis of diseases, Treatment of diseases (answer any two, each carries 1 mark)
8. Alpha helix- peptide chain forms regular helical coils (1 mark)
- coils are stabilized by hydrogen bond
Beta pleated sheet- sheet like structure (1 mark)
9. Essential amino acids- cannot be synthesized by the body (1 mark)
Non- essential amino acids- can synthesized in the body. (1 mark)
10. Hydroxylation in liver (1 mark)
Hydroxylation in kidney (1 mark)

Part C

Short Essay Type Questions
Answer any four questions
Each question carries five marks

11.Heteropolysacharides- definition, example (2 marks)
Homopolysacharides- definition, example (2marks)
Structure (1 mark)
12.Defense proteins (examples)
Contractile proteins (examples)
Transport proteins (examples)
Structural proteins (examples)
Enzymic proteins (examples)
Nutrient and storage proteins (examples)
Regulatory proteins (examples)
Toxic proteins (examples) (answer any five, each carries one mark)
13. Saturated fatty acids (examples) (1 mark)
Unsaturated fatty acids – MUFA, PUFA, (examples)(1 mark)
Short chain fatty acids, medium chain fatty acids, long chain fatty acids, very long chain fatty acids (examples)(3 marks)
14. Rhodopsin (2 marks)
Conopsin (2 marks)
Retinal (1 marks)
15. ATP production (1 mark)
Coenzyme in energy metabolism (1 mark)
Pentose Phosphate Pathway (1 mark)
Nervous system support (1 mark)
Amino acid metabolism (1 mark)
16.Structure (3 marks)
Water soluble (1 mark)
Sensitive to heat, light and oxidation (1 mark)

Part D

Long Essay Type Questions
Answer any two questions
Each question carries ten marks

17.Definition (1 mark)
Chemical classification: with examples (5 marks)
Monosaccharide
Oligosaccharides
Polysaccharides
Physiological Classification with examples (4 marks)
Simple
Complex

18. Saturated & Unsaturated fatty acids(6 marks)
Omega 3 fatty acids, omega 6 fatty acids, omega 9 fatty acids.(3 marks)
Short chain, medium chain fatty acids, long chain fatty acids, very long fatty acids(4 marks)

19. Structure (1 mark)
Vitamin D2 and Vitamin D3 (2 marks)
Functions- (5 marks)
calcium & phosphorus metabolism
Bone health
Cell growth and differentiation
Immune system regulation
Muscle function
Hormonal Balance
Properties-(2 marks)
Solubility
Stability
Bioavailability
Sunlight Synthesis

20. Structure (1 mark)
Functions- (6 marks)
Collagen production
Antioxidant defense
Iron absorption
Immune system
Wound healing
Neurotransmitter production
Properties- (3 marks)
Solubility
Stability
Antioxidant
Taste